Ram Certificates and Quantitative Closure for Positively Curved Einstein Four-Manifolds

M671: Source Audit, Boundary-Sensitive Algebra, and Conditional Residual Bounds

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Abstract

We examine the pinned manuscript *Positively curved Einstein four-manifolds*, including its analytic and topology appendices, against actual local Ram corpus passages on Bernstein enclosures, residual accounting, physical quadratic forms, and exact semidefinite faces. The source claims classification without orientability assumptions; that headline theorem is not independently established here. We separate its differential, global, finite-algebraic, and equality-case obligations. Exact replay of forty coefficient groups supports explicit boundary-vanishing matrix and scalar margins. We prove their transfer to an additional quadratic gradient defect and a common two-well potential, which shortens the final rigidity argument. The rational comparison deficit also controls fourth powers of gradients of transformed Weyl norms, and a strict spectral margin supplies explicit positive obstructions to two nonzero blocks under small total error. Moment imbalance has a precisely charged variance cost, and the common potential gives conditional spectral-concentration estimates. All extensions retain the stated scalar and tensor inputs; none asserts stability for arbitrary approximate Einstein metrics or a nonnegative-curvature classification. Classical positivity and geometric techniques are distinguished from private Ram representations, and computational checks are finite and local. Contribution assessment: C3/5, exact boundary margins yield a shorter terminal proof and strengthened quantitative residual closure while the global geometric chain remains an independent obligation.

Confidential research report. Private corpus passages were read locally and were not submitted in public searches. The report date is October 7, 2026; the verification session crosses into October 8 UTC.

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1 Pinned Problem and Logical Scope

The catalogue entry is manuscript 671, family 348, applicability level L5. Applicability is not a validity judgment or a contribution score. The source is the local snapshot of OpenAI's September 23, 2026 manuscript [1], pinned to

`adc7f1241b42e322a6451854ab7e4b4c146bf78a`.

The local directory, relative to the study root, is

`_sources/openai-math/preprints/Positively-curved-Einstein-four-manifolds-September-23-2026/`.

The catalogue pin identifies the snapshot being investigated; reading local files is not cryptographic authentication of the remote Git history. File hashes and read scope are recorded separately.

The source's quantifiers are: for every connected smooth closed Riemannian four-manifold (M, g), without an orientability assumption, if $\mathrm{Ric}_g = \lambda g$ for a constant λ and every tangent two-plane has strictly positive sectional curvature, then some $a > 0$ makes (M, ag) isometric to a round S^4, Fubini–Study $\mathbb{CP}^2$, or round $\mathbb{RP}^4$. Closed means compact with empty boundary. This is an isometric classification up to homothety, not merely a homeomorphism classification.

Strict positivity implies $\lambda > 0$. For $g' = (\lambda/3)g$ one has $\mathrm{Ric}_{g'} = 3g'$. All formulas below use this normalization. An orientation is chosen for the main analysis; the orientation double cover is used only at the last stage. The normalized round sphere has sectional curvature one. The projective model is twice the metric obtained from the unit Hopf sphere S^5; its sectional curvature ranges from $1/2$ to 2 and its Ricci tensor is $3g$.

Three statement classes. Statements explicitly proved below are local algebraic or analytic implications. Statements designated *source inputs* describe the manuscript's argument without asserting independent verification of its complete proof. A conditional conclusion requires all its displayed hypotheses; a coefficient certificate alone never supplies those hypotheses. No theorem from the nonnegative/equality-boundary companion M669 is used to infer the strictly positive M671 theorem.

2 Spaces, Normalizations, and Boundary Conventions

Let $d\nu = d\mathrm{vol}_g/\mathrm{Vol}(M)$ and $\mathbf{E}f = \int_M f\, d\nu$. On the oriented manifold, $\Lambda^2 = \Lambda^+ \oplus \Lambda^-$ and

$$\mathcal{R}_{\Lambda^2} = \mathrm{diag}(\mathrm{Id} + T, \mathrm{Id} + U), \qquad T \in \mathrm{Sym}_0(\Lambda^+), \quad U \in \mathrm{Sym}_0(\Lambda^-).$$

Both blocks have rank-three fibers, are real symmetric and trace-free, and use Hilbert–Schmidt matrix norms. Set

$$v = |T|/\sqrt{6}, \quad b = |U|/\sqrt{6}, \quad x = -\lambda_{\min}(T), \quad y = -\lambda_{\min}(U), \quad p = \nabla v, \quad q = \nabla b.$$

For a unit simple two-form its two Hodge components both have norm $1/\sqrt{2}$, and conversely every such pair is simple. Minimizing the two block quadratic forms independently gives

$$K_{\min} = 1 - (x + y)/2, \qquad \sec > 0 \iff x + y < 2, \qquad v \le x \le 2v, \quad b \le y \le 2b. \tag{1}$$

For the last bounds, write the eigenvalues as $(-x, a, x - a)$ with $-x \le a \le 2x$, so $3x^2/2 \le |T|^2 \le 6x^2$. In particular $v + b < 2$, but this weaker norm inequality does not replace $x + y < 2$ in the Hessian positivity argument.

The polynomial certificates are evaluated on the closed triangles

$$\mathcal{D} = \{v, b \geq 0 : v + b \leq 2\}, \qquad \mathcal{D}_+ = \mathcal{D} \cap \{v \geq b\}.$$

These are algebraic domains containing boundary points that cannot all occur in the standing strict geometry. Write

$$\delta = 2 - v - b, \quad K = 4 + v^2 + b^2, \quad e = |\nabla T|^2/6, \quad \bar{e} = |\nabla U|^2/6.$$

The norms are Lipschitz; their gradients are weak gradients, equal to their classical gradients away from zeros. On a zero set the corresponding tensor and scalar derivatives vanish almost everywhere. A quotient such as e/v is assigned zero there only after the integrability argument below, not by cancellation of an undefined expression.

The Laplacian is $\Delta = \operatorname{tr} \nabla^2$, so $\mathbf{E}(f\Delta f) = -\mathbf{E}|df|^2$. The tensor convention is $[\nabla_i, \nabla_j]\xi_k = R_{ijka}\xi_a$, with $R_{ijij} > 0$ on a positively curved orthonormal plane. All covariant derivatives are taken before choosing a pointwise eigenbasis; repeated eigenvalues do not license differentiation of a chosen eigenframe.

3 Stepwise Source Dependency Audit

Every substantive section in `build/sections/01-introduction.tex` through `08-classification.tex`, both appendices, the two figures, main file, references, and verification documentation was read. The source references below are literal labels in those files, not routing-map references.

3.1 Curvature algebra and weak scalar identities

Section 2, `prop:weyl-derivative-equations`, claims

$$\Delta T = 6(T - N), \quad N = (T^2)_0 = T^2 - 2v^2 \operatorname{Id}, \qquad \Delta Z = (13 - 20Q_{5/2}(T))Z, \quad Z = \nabla T.$$

Its gradient bundle has representation type $(5/2, 1/2)$ and its projection is parallel. These are oriented-frame representation types, not an assumption that M is spin. The Bianchi parametrization, `lem:gradient-parametrization`, has three arbitrary tangent vectors (p, k, l) and physical energy

$$e = \frac{5}{3}|p|^2 + 5|k|^2 + |l|^2. \tag{2}$$

For normalized eigenvalues $s_i = t_i/v$, define $\sigma = \sum_i s_i^3/6$ and $\rho = \sum_i s_i^2(s_j - s_k)/(6\sqrt{3})$ cyclically. Then $\sigma^2 + \rho^2 = 1$ and, for $r_0 = x/v \in [1, 2]$, $\sigma = (3r_0 - r_0^3)/2$. Only products with adequate powers of v extend across zeros.

The source's `prop:weak-norm-equation` states the distributional equality

$$\Delta v = 6v(1 - \sigma v) + (e - |p|^2)/v, \qquad e/v \in L^1(d\nu). \tag{3}$$

The regularization $v_\varepsilon = (v^2 + \varepsilon^2)^{1/2}$ has derivative term

$$q_\varepsilon = e/v_\varepsilon - v^2|p|^2/v_\varepsilon^3, \qquad \frac{2}{5}e/v_\varepsilon \leq q_\varepsilon \leq e/v_\varepsilon.$$

Integrating the smooth equation bounds $\mathbf{E}(e/v_\varepsilon)$; monotone and dominated convergence give an L^1 limit. This is the essential exclusion of a hidden singular measure on $\{v = 0\}$. Uniform and H^1 smooth approximation then permits bounded Lipschitz tests. These steps must precede the multiplier and quartic integrations.

3.2 Scalar gaps, heat identities, and balanced moments

Section 3 uses the strengthened variance inequality

$$\mathrm{Var}(f) \le \mathbf{E}\frac{|\nabla f|^2}{4+v^2+b^2}, \qquad f \in H^1(M;\mathbb{R}), \tag{4}$$

derived by pairing the curvature action on $\mathrm{Hess}_0\phi$ with a mean-zero Poisson solution. The operator upper bound is obtained from $x+y<2$; the invariant contraction supplies $3(v^2+b^2)|\nabla\phi|^2$. The constant four in the denominator must not be lost in normalization.

The sharp normalized Sobolev inequality is

$$(\mathbf{E}|f|^4)^{1/2} \le \mathbf{E}(f^2+|df|^2/2), \qquad f \in H^1.$$

Testing with $(v^2+\varepsilon^2)^{1/6}$ and retaining the refined Kato coefficient in (2) yields the classical Weyl gap

$$T \not\equiv 0 \Longrightarrow \mathbf{E}v^2 \ge 1. \tag{5}$$

Equality entails $v=\sigma=1$ and $\nabla T=0$. The proof uses equality in Cauchy–Schwarz and connectedness; it does not divide by a possibly vanishing norm. The same assertions hold for U.

On the oriented manifold, $\mathrm{Ric}=3g>0$ and the Bochner identity for harmonic one-forms give $b_1=0$; Poincare duality then gives $b_3=0$ and $b_0=b_4=1$. With $n_\pm$ the dimensions of harmonic $\Lambda^\pm$ two-forms, the characteristic identities are

$$\begin{aligned} \chi &= 2+n_++n_-, & \tau &= n_+-n_-, \\ 8\pi^2\chi &= 6V_M(1+s_0), & 12\pi^2\tau &= 6V_M\mathbf{E}(v^2-b^2), \qquad s_0=\mathbf{E}(v^2+b^2). \end{aligned} \tag{6}$$

Here $U=0$ implies $n_-=0$ by the Hodge potential $4\,\mathrm{Id}-2U$; this is not a topological inference from a vanishing scalar average alone.

Section 4 first claims $V_M/(8\pi^2/3) \le 1-3s_0/40$. For two nonzero blocks, $2 \le s_0 < 4$, and the characteristic formulas give

$$\chi \le 2(1-3s_0/40)(1+s_0) < 7, \qquad |\tau|/\chi < 4/15.$$

Parity leaves only $(\chi,|\tau|)=(5,1)$ if $\tau \neq 0$. With $P=\mathbf{E}b^2$ the smaller moment, this alternative implies

$$\mathbf{E}v^2 = (13P+3)/7, \quad P \ge 1, \quad 28\mathbf{E}b+6P \le 25.$$

The constrained probability-measure obstruction is genuinely additional. Its function

$$H(z)=\sqrt{z}(1+z^2/40-z^4/1152), \quad \Phi(z)=16H(z)^2+9z(1-z), \quad z\in[0,2]$$

must satisfy $16(\int H)^2-\int\Phi<0$ for every feasible probability measure. The source linearizes at a hypothetical nonnegative maximizer, uses strict feasibility for separation, and proves the support lies at one interior point and 2 using a positive third derivative. It then checks two endpoint weights by degree-22 polynomials. This is not a generic assertion that two constraints always give a two-point maximizer. The derivative signs and endpoint polynomials are essential. Combined with (4) and (3), it contradicts a nonnegative value of that functional and forces $\tau=0$, hence $\mathbf{E}v^2=\mathbf{E}b^2$.

3.3 The two radial-volume arguments

The upper estimate uses a parallel normal Jacobi matrix $\mathcal{R}(t)$ along geodesics, with $0 < \mathcal{R} < 3\,\mathrm{Id}$ and $\operatorname{tr}\mathcal{R} = 3$. Let $A = \mathcal{R} - \mathrm{Id}$ and $D(t) = \det \mathcal{J}(t)$ for $\mathcal{J}'' + \mathcal{R}\mathcal{J} = 0$, $\mathcal{J}(0) = 0$, $\mathcal{J}'(0) = \mathrm{Id}$. Invariant unit-tangent-bundle expectation $\mathcal{E}$ obeys

$$\mathcal{E}|A|^2 = \tfrac{3}{2}s_0, \qquad \mathcal{E}|A'|^2 \le \tfrac{3}{2}\mathcal{E}|A|^2.$$

The discrete block-tridiagonal Dirichlet matrix has diagonal $2\,\mathrm{Id} - a^2\mathcal{R}(ia)$ and adjacent blocks $-\mathrm{Id}$. Its determinant times a^3 converges to $D(t)$. Preconjugate positivity is established before using Schur complements. The first sine-mode compression gives, for $t < \pi$ before conjugacy,

$$D(t) \le \sin^3 t \det(\mathrm{Id} - \beta B), \quad \beta = t^2/(\pi^2 - t^2), \quad B = \frac{2}{t}\int_0^t A(s)\sin^2(\pi s/t)\,ds.$$

The upper polar integral uses D_c, set to zero after first conjugacy; it allows multiplicity and does not identify cut points with conjugate points. Rational sine minorants give a coefficient exceeding $2/25$, from which the weaker $3/40$ is used.

Section 8's lower estimate is separate and conditional on $s_0 = 3$. The injectivity bound permits integration only to $\pi/\sqrt{3}$. With a positive discrete Green matrix G, ordered products of curvature matrices give

$$\mathcal{E}\operatorname{tr} H_N^k \le \tfrac{5}{6}3^k \operatorname{tr} G^k \quad (k \ge 2), \qquad \operatorname{tr} H_N = 3\operatorname{tr} G.$$

No commutation or independence assumption is permissible. Jensen and the scalar determinant limit yield

$$\mathcal{E}D(t) \ge t^3 \left(\frac{\sin(\sqrt{3}t)}{\sqrt{3}t}\right)^{5/6} e^{-t^2/12}.$$

The logarithmic estimates are uniform at each fixed interior time, not at the singular endpoint. Endpoint passage uses nonnegative integrands and monotone convergence. Rational integrals produce $V_M/\pi^2 > 1343/1000 > 4/3$. These are global analytic implications in addition to the finite integrals.

3.4 Weighted Hessian and first-derivative coupling

Section 5 isolates the full, generally nonsymmetric Hessian:

$$\nabla Z = Y + D_0, \quad Y \text{ of type } (3,1), \quad D_0 = \tfrac{5}{12}\mathrm{Tr}^*\Delta T, \quad |D_0|^2 = \tfrac{5}{12}|\Delta T|^2.$$

The trace component contains the curvature commutator and cannot be replaced by a symmetric trace. The compressed operator

$$\mathcal{P} = 4 - \tfrac{5}{6}(T_D + 2B_F(T)) - U_D$$

preserves the combined types and on Y has lower bound $4 - 2x - 2y > 0$. Its individual summands need not preserve those types. $\mathcal{P}$ is not parallel, and the integrated calculation retains the derivative of the matrix-slot action $B_F(T)$.

For every global Lipschitz weight $m \ge m_0 > 0$, the source establishes

$$0 \le \mathbf{E}\left[mG_v - \frac{5}{6}\sum_i m_i \frac{\operatorname{tr}(Z_i \sum_j Z_j^2)}{6} + \frac{\langle \mathcal{P}(dm \otimes Z)_Y, (dm \otimes Z)_Y\rangle}{24m}\right], \tag{7}$$

where

$$G_v = (-42 - 3b^2 - 10v^2/3)e + 75B + 25C_N/3 - 100E_B/3 + 5v^2|p|^2.$$

Here $B = \langle B_F(T)Z, Z\rangle/6$, $C_N = \langle B_F(N)Z, Z\rangle/6$, and $E_B = |Q_{5/2}(T)Z|^2/6$. In the coordinates (2), the latter is represented by $\mathsf{B}\,\mathrm{diag}(3/5, 1/5, 1)\mathsf{B}$, not B^2. This is a direct physical-metric correspondence with the private corpus.

Section 6 fixes $M = 1 - 3(v - b)/8$, $m = M^2$, and exchanges these for U. On $\mathcal{D}$, $1/4 \le M \le 7/4$. It adds a nonnegative Hessian identity, a balanced variance expression, and two zero-integral weak-norm tests. Their coefficients are

$$\begin{gathered}
D = 158, \quad h = 17/10, \quad c = 2/5 - 3vb/10 \ge 1/10,\\
a = v - 1, \quad s = b - 1, \quad n = \delta/2, \quad t = (v - b)/2,\\
\kappa = -\tfrac{D}{6}(a - s) + z_1a^2 + (60 - 2D/9 + 2z_1)as + z_2s^2 + \sum_{i=0}^{3} z_{3+i}a^i s^{3-i} + \sum_{i=0}^{4} z_{7+i}a^i s^{4-i},\\
4(z_1, \dots, z_{11}) = (-43, -45, -63, 21, 183, 208, 61, 46, 101, 79, -41),\\
4\Theta = n^2(1 - n)(105 + 5n) + t^2(4 + 360n - 586n^2) - 4t^4,\\
R(v) = \frac{4(1 + v)}{1 + 6v + v^2}, \quad r = R(v)|p|^2 + R(b)|q|^2, \quad j = (|p|^2 + |q|^2)(|p|^2/v + |q|^2/b).
\end{gathered} \tag{8}$$

The resulting $\mathcal{I} = \mathcal{I}_4 + \mathcal{I}_2 + \mathcal{I}_0$ has nonnegative expectation in the balanced case. Its proposed bounds are

$$\mathcal{I}_4 \le -cj, \quad \mathcal{I}_2 \le 2\Theta r, \quad \mathcal{I}_0 \le -\Theta^2/c, \quad \mathcal{I} \le -\frac{(\Theta - cr)^2}{c} - c(j - r^2). \tag{9}$$

The quartic comparison first removes nonnegative Kato defects and then makes a relaxed spectral comparison; the relaxed pair need not be geometrically realized. The quadratic comparison uses a Schur complement and monotonicity in σ, with a separate $\sigma = 1$ argument to avoid dividing by $1 - \sigma$. The potential uses the ordered domain and exchanges globally defined labels only after differentiating. Zero sets are reached by setting the zero block's derivative parameters to zero before taking an admissible axial limit. These order-of-operation restrictions are mathematical obligations, not implementation preferences.

3.5 Finite certificates and geometric completion

Section 7 converts rational polynomials to interval and triangular Bernstein coefficients. It uses two matrix arrays $\mathcal{A}$ and $\mathcal{B}$ for the shape and axial comparisons. The first is degree eight on three triangles covering $\mathcal{D}$; the second uses degrees nine and ten on six triangles covering $\mathcal{D}_+$. Scalar certificates cover quartic signs, two boundary potentials, and an even potential written in (n, t) coordinates. A positive determinant of the polynomial matrix is not the same assertion as a positive determinant of every Bernstein coefficient matrix. The source uses the latter sufficient condition together with nonnegative diagonals and exact zero exceptions.

Section 8's active-gradient argument uses $R(v) \le v^{-1/2}$, equality only at $v = 1$, and the fact that a Lipschitz function has zero gradient almost everywhere on each level set. Thus (9), its integral sign, and the norm gaps would exclude two nonzero blocks. Our quantitative replacement is proved in Section 6.

After $U = 0$ with $T \not\equiv 0$, the source's one-block alternatives are

$$V_M/\pi^2 = (8 - 2n_+)/3, \qquad s_0 = 3n_+/(4 - n_+), \qquad n_+ = 1 \text{ or } 2.$$

The $n_+ = 2$ case would give $s_0 = 3$, $V_M = 4\pi^2/3$ and is excluded by the conditional lower comparison. The surviving equality case has parallel T with eigenvalues $(2, -1, -1)$. Tensor matching, simple connectivity, and continuation of isometry germs identify the metric. Alternatively, the classical half-conformally-flat classification can supply this last step once its hypotheses are established; it does not supply the preceding one-block vanishing.

For a nonorientable M, its connected orientation double cover is one of the oriented models only after the oriented result. An orientation-reversing diffeomorphism of $\mathbb{CP}^2$ is excluded by its cohomology ring, since $h \mapsto \pm h$ fixes h^2. A free orthogonal involution on $S^4 \subset \mathbb{R}^5$ has no $+1$ eigenspace and therefore is $-\operatorname{Id}$, producing $\mathbb{RP}^4$. There is no unproved enumeration of arbitrary spherical quotient groups in this double-cover argument.

3.6 Appendix dependencies and remaining verification boundary

Appendix A establishes Poisson solvability on mean-zero H^1, weak-to-strong elliptic regularity, and density; its subcritical Sobolev minimizer is made positive before a negative power is taken. Compactness is used below exponent four, followed by a limit to four. Its form complex $\Lambda^0 \to \Lambda^1 \to \Lambda^+$ has potentials 0, $3\operatorname{Id}$, $4\operatorname{Id} - 2T$. The heat-supertrace argument cancels signed rank and signed potential, leaving

$$(16\pi^2)^{-1}(6 + 10v^2 + 2b^2).$$

The parametrix remainder, spectral trace, Hodge identification, and the local coefficient contraction remain analytic steps beyond polynomial positivity.

For the oriented manifold, Appendix B derives diameter $\leq \pi$, injectivity radius $\geq \pi/\sqrt{3}$, and simple connectivity from the index form and orientation-preserving normal holonomy in dimension three. The injectivity conclusion is not applied directly to $\mathbb{RP}^4$. The appendix distinguishes interior conjugacy from a conjugate endpoint, obtains two minimizing branches at a sub-conjugate cut point, and uses simultaneous endpoint shortening. The normal periodic parallel field is available because the normal holonomy lies in $\mathrm{SO}(3)$. These are the precise global inputs for the two polar integrations and final continuation.

The audit does not constitute a proof-assistant formalization or an independent verification of every representation-theoretic, heat, Jacobi, and topology step. In particular, a successful coefficient replay cannot repair an incorrect tensor collection identity or a missing global hypothesis. The local source copy contains the verification README and coverage note but not the advertised `verify.py`, `polynomial.py`, or `results.json`; the advertised packaged test is not reported as executed.

4 Actual Ram Correspondences

The following are actual local passages, read at their original paths. They supply representations and certification disciplines; they do not by themselves contain M671's geometric theorem.

1. `C:/Users/mural/OneDrive/Documents/RamCalBooks/volumes/volume16/chapters/ch05d-clearance.tex`, sections *Polynomial enclosures in Bernstein form* and *Subdivision without introducing approximation*, labels `v16c05:bernstein`, `v16c05:conversionproof`, `v16c05:hull`, `v16c05:hullbound`. The power-to-Bernstein identity and convex-hull proof match M671's interval certificate exactly. The preceding between-knot counterexamples explain why finite sampling is not a continuum sign proof. Its strict-positive subdivision stopping rule explicitly does not cover touching zeros.

2. `C:/Users/mural/OneDrive/Documents/My Documents 2026/res-rt/res-math/TR-item-047-astra.tex`, sections *Algebraic and Moment Reformulation*, *Ram Analysis and Synthesis*

Panels, *Global Optimality Sandwich*, *Making the Moment and Identity Tests Exact*, and *Residual-Aware SOS Lower Bounds*, label `sec:soserror047`. Its exact polynomial identity plus Gram-sign requirement corresponds to the finite layer; its inexact Gram and identity theorem charges the residual instead of identifying a small interval with zero. In M671 that principle becomes the explicit integral defect in Section 6. No generic moment relaxation is used to assert that the constrained two-point measure argument is solved.

3. `C:/Users/mural/OneDrive/Documents/My Documents 2026/res-rt/res-math/TR-item-305-astra.tex`, *The actual Ram bridge*, *Rational facial certificates and their exact scope*, *Quantitative exposure: leakage is not exact rank*, *Affine repair and conditional rational bit bounds*, and the weak-infeasibility and irrational-feasibility sections; labels `thm:face`, `thm:repair`, `thm:weak`, `thm:irrational`. The physical metric distinction explains the $\mathsf{B}H^{-1}\mathsf{B}$ expression above. Exact zero coefficient matrices and vanishing jets must be retained as exact constraints. Approximate PSD does not establish a face, and rational discovery is not guaranteed in a general SDP.

Bernstein expansions, convex hulls, Schur complements, Weyl perturbation estimates, and facial reduction are classical. Calling their finite data a Ram analysis/synthesis representation records an organizational and coordinate choice; it does not claim historical discovery of those techniques. The realized local contributions are the specialized boundary estimates and residual-to-variance consequence proved next. Neither a measured runtime gain nor a general algorithmic complexity improvement is claimed.

5 Boundary-Preserving Matrix Certificates

Let $\lambda = (\lambda_1, \lambda_2, \lambda_3)$ be barycentric coordinates on a closed nondegenerate triangle. For $|\alpha| = N$, write $B_\alpha^N = \binom{N}{\alpha}\lambda^\alpha$. These nonnegative weights sum to one. An entrywise Bernstein expansion of a symmetric matrix polynomial is an exact identity $P = \sum_\alpha C_\alpha B_\alpha^N$.

Theorem 5.1 (Active-coefficient margin and exact zeros)**.** *Suppose every C_α is either exactly zero or positive definite, and let S be the nonzero indices. If $C_\alpha \succeq \gamma_\alpha I$ with rational $\gamma_\alpha > 0$, then*

$$P(\lambda) \succeq \Gamma(\lambda) I, \quad \Gamma = \sum_{\alpha \in S} \gamma_\alpha B_\alpha^N \geq \gamma W, \quad \gamma = \min_S \gamma_\alpha, \quad W = \sum_S B_\alpha^N.$$

If symmetric coefficient perturbations satisfy $E_\alpha = 0$ outside S and $\|E_\alpha\|_2 \leq \theta\gamma_\alpha$ on S for $0 \leq \theta < 1$, then

$$P + \sum E_\alpha B_\alpha^N \succeq (1 - \theta)\Gamma I.$$

For a rational positive definite 2×2 coefficient, one may take $\gamma_\alpha = \det C_\alpha / \operatorname{tr} C_\alpha$.

Proof. Sum the quadratic-form inequalities with the nonnegative Bernstein weights. The perturbation satisfies $E_\alpha \succeq -\theta\gamma_\alpha I$. For the last assertion, if $0 < \lambda_1 \leq \lambda_2$ are the eigenvalues, then $\lambda_1 = \det C / \lambda_2 \geq \det C / \operatorname{tr} C$. All steps are valid on the triangle's edges as well as its interior. □

The theorem is a supplied-certificate verifier, not a guarantee that arbitrary positive matrix polynomials have this representation at the chosen degree. Lower determinant bounds alone do not give γ without an upper trace bound; exact arrays provide both. Negative Bernstein coefficients are inconclusive, not pointwise counterexamples.

Proposition 5.2 (The two M671 boundary profiles)**.** *Assume the source's exact nonzero/zero coefficient pattern for $\mathcal{B}$. On triangle ACE, of degree ten, its only zero is the coefficient at C, and*

$$\mathcal{B} \succeq \gamma_{ACE}(1-\lambda_C^{10})I \succeq \gamma_{ACE}(1-\lambda_C)I.$$

On triangle CBE, of degree ten, the zeros are $(10,0,0)$ and $(9,1,0)$ in the ordered (C,B,E) coordinates, and

$$\mathcal{B} \succeq \gamma_{CBE}(1-\lambda_C^{10}-10\lambda_C^9\lambda_B)I \succeq \gamma_{CBE}(\lambda_E+\lambda_B^2)I. \tag{10}$$

Here each γ is the minimum determinant-to-trace bound over that triangle's nonzero coefficients. Both inequalities are conditional solely on those finite coefficient signs and identities, not on geometric realizability of (v,b).

Proof. The first assertion follows by subtracting the sole missing basis weight. For the second, interpret the multinomial weights as probabilities for ten independent draws with probabilities $(\lambda_C,\lambda_B,\lambda_E)$. The missing outcomes are all C, or exactly one B and nine C. Every outcome with its first draw E, and every outcome with its first two draws B, is active. Those two events are disjoint and have probabilities λ_E and λ_B^2. This proves the lower bound. The argument is a finite polynomial identity/inequality, not a stochastic assumption about curvature. □

The coordinates of these vertices are $A=(1/2,1/2)$, $C=(1,1)$, $B=(3/2,1/2)$, and $E=(1,1/2)$. Thus on CBE, $\lambda_E=2\delta$ and $\lambda_B=2(v-1)$. The lower profile is $2\delta+4(v-1)^2$. It records first-order inward and second-order tangential protection at the singular boundary point; no uniform positive eigenvalue exists there. On ACE, $1-\lambda_C\geq\delta$, giving an inward bound.

Exact zero preservation matters. The toy scalar u^2 on $[0,1]$ is nonnegative but $u^2-\varepsilon$ fails at $u=0$ for every $\varepsilon>0$. Analogously, adding $-\varepsilon I$ at the C coefficient cannot be justified by the nonzero margins. Theorem 5.1 permits controlled perturbations of active coefficients while holding the exact boundary constraints; it is not tolerance-based PSD acceptance.

Return to the source's rational matrix. For $v,b>0$, let $\mathcal{I}_{2,\mathrm{ax}}$ be the source's axial upper comparison, so $\mathcal{I}_2\leq\mathcal{I}_{2,\mathrm{ax}}$. The matrix representing $2\Theta r-\mathcal{I}_{2,\mathrm{ax}}$, not necessarily $2\Theta r-\mathcal{I}_2$, has diagonal entries $H_v/(vKr_v)$, $H_b/(bKr_b)$ and off-diagonal $-J$, where $r_v=1+6v+v^2$ and $r_b=1+6b+b^2$. To specify all its polynomial inputs, use (3.4), $\bar\kappa=\kappa(b,v)$, and

$$\begin{aligned}
X_v &= m(-70+100v-25v^2-5b^2)+\tfrac{5}{3}h\{6((v-1)^2+(b-1)^2)+32(v-1)\delta\},\\
H_v &= 8vK\Theta(1+v)-r_v\{vD+K[v(X_v+\partial_v\kappa)+2\kappa/3]\},\\
H_b &= H_v(b,v), \qquad J=\tfrac{1}{2}(\partial_b\kappa+\partial_v\bar\kappa)-D/K.
\end{aligned}$$

The source's polynomial matrix is (H_v,H_b,S_*KJ), with

$$S_*=(\alpha br_v+\alpha^{-1}vr_b)/2, \qquad \alpha=5/4.$$

For $vb>0$, $S_*\geq\sqrt{vbr_vr_b}$, so its PSD determinant inequality suffices for this axial difference matrix, irrespective of the sign of J. Dividing by v or b at a zero block is not justified by that statement; one instead uses the compatible zero-derivative limit already specified in the source audit.

5.1 Replayed vanishing-factor margins

Put $L_b = 10h\delta b - \bar{\kappa}$, $L_v = 10h\delta v - \kappa$, and

$$\begin{aligned} E_0 &= D(1-\delta^2/4)(v-b)^2 + 30h\delta\{v^2(v-1)^2 + b^2(b-1)^2\} \\ &\quad + 6\{v(v-1)\kappa + b(b-1)\bar{\kappa}\}, \\ F_0 &= cE_0 - \Theta^2, \qquad G_0 = F_0 + 12c\delta^2 L_b. \end{aligned}$$

Here F_0 is the source's potential polynomial F; it is distinct from the one-variable transform $F(s)$ used below.

Proposition 5.3 (Explicit polynomial margins)**.** *On the full closed ordered triangle $\mathcal{D}_+$,*

$$\mathcal{B} \succeq \tfrac{1}{10}(1-b)^2 I_2, \qquad G_0 \geq \tfrac{2}{5}(1-b)^4. \tag{11}$$

Proof. For an ordered triangle with vertices $p_i = (v_i, b_i)$ and multi-index $|\alpha| = N$, the Bernstein coefficient of $(1-b)^k$ is

$$w_\alpha^{(k)} = \binom{N}{k}^{-1} \sum_{\substack{|\beta|=k \\ \beta \leq \alpha}} \prod_{i=1}^{3} \binom{\alpha_i}{\beta_i} (1-b_i)^{\beta_i}. \tag{12}$$

This follows by multiplying the degree-k multinomial expansion by $(\lambda_1+\lambda_2+\lambda_3)^{N-k}$ and comparing normalized coefficients. All weights are nonnegative since $b_i \leq 1$. They vanish at every exceptional zero coefficient of the corresponding source object.

Apply the source's exact triangular conversion to the displayed entries of $\mathcal{B}$ and G_0. For nonzero matrix coefficients C_α, the following table records

$$\min_{w_\alpha^{(2)}>0} \frac{\det C_\alpha}{\operatorname{tr} C_\alpha \, w_\alpha^{(2)}};$$

for the scalar rows it records $\min_{w_\alpha^{(4)}>0} g_\alpha / w_\alpha^{(4)}$.

Object	Triangle	Degree	Exact minimum
$\mathcal{B}$	OAS	9	43735879/6451650
	ASE	9	32821379939188729/4998273275750400
	ACE	10	161755713049/460285848000
	SBE	9	43735879/6451650
	CBE	10	7166179946253733/2289425685120000
	SUB	9	865893637/554495625
G_0	OSC	9	170363/5120
	SCU	9	2497/5040

The remaining vertices are $O = (0,0)$, $S = (1,0)$, $U = (2,0)$. The six matrix triangles cover $\mathcal{D}_+$, as do the two scalar triangles. Every coefficient is obtained by the explicit finite formula

$$[p]_{ij} = \sum_{d \leq i,\, e \leq j} \frac{\binom{i}{d}\binom{j}{e}}{\binom{N}{d+e}\binom{d+e}{d}} P_{de}, \quad P(z,w) = p(p_3 + (p_1-p_3)z + (p_2-p_3)w).$$

The exact replay computes these arrays over $\mathbb{Q}$; the local check independently verifies their shifted PSD signs and weight identities from the stored rational entries. The minimum matrix ratio is $161755713049/460285848000 > 1/10$ and the minimum scalar ratio is $2497/5040 > 2/5$. Coefficients whose weight is zero retain their original nonnegative sign. Summing proves (11), including every edge. This is a finite coefficient proof, not a sampled eigenvalue test. □

Theorem 5.4 (Additional quadratic gradient defect)**.** *Assume the source's shape-to-axial comparison for $\mathcal{I}_2$. With the explicit coefficients (3.4), its quadratic inequality strengthens to*

$$\mathcal{I}_2 \le 2\Theta r - \frac{(1-\min(v,b))^2}{2720}(|\nabla v|^2 + |\nabla b|^2). \tag{13}$$

It holds almost everywhere for the source's strict geometry, including zero norms with the specified zero-derivative convention.

Proof. First assume $v \ge b > 0$. Congruence of (11) by

$$L = \operatorname{diag}((vKr_v)^{-1/2}, (bKr_b)^{-1/2})$$

gives $L\mathcal{B}L \succeq (1-b)^2 I/2720$, because $vKr_v, bKr_b \le 2 \cdot 8 \cdot 17 = 272$ on $\mathcal{D}$. Its diagonals are exactly those of the axial difference matrix, but the absolute off-diagonal is

$$S_*|J|/\sqrt{vbr_v r_b} \ge |J|.$$

For fixed diagonals a, d, the smaller eigenvalue of a real symmetric two-by-two matrix with off-diagonal o is

$$\tfrac{1}{2}\{a + d - \sqrt{(a-d)^2 + 4o^2}\},$$

which decreases as $|o|$ increases. Thus the axial difference matrix also has smallest eigenvalue at least $(1-b)^2/2720$. This uses the eigenvalue formula, not a false Loewner comparison obtained by changing only off-diagonal entries. Apply the bound to each tangent component of (p, q), and then the source's shape comparison.

For $v < b$, exchange the algebraic labels and the vector entries. No derivative of $\min(v, b)$ is taken. If one block vanishes, set its derivative parameters to zero first and approximate its spectrum by small axial spectra while holding the opposite block and its derivatives fixed; strict sectional positivity leaves room for this limit. The surviving diagonal rational expression has a positive denominator. The subtracted margin is continuous and tends to $(|p|^2 + |q|^2)/2720$ because the smaller norm tends to zero. If both blocks vanish, both norm gradients vanish almost everywhere. This proves the claimed extension. □

The scalar margin in (11) cannot by itself be subtracted indiscriminately from the potential: the source uses G_0 only in the branch $L_b < 0$, whereas in the branch $L_b \ge 0$ it uses F_0. A further bound for F_0, established next, is what permits a common gain.

5.2 A two-well potential and shorter terminal argument

Proposition 5.5 (A common potential remainder)**.** *On $\mathcal{D}$, put $n = (2 - v - b)/2$, $t = (v - b)/2$ and*

$$\mathcal{H}(n,t) = n^4(1-n) + t^4, \qquad 0 \le n \le 1, \quad |t| \le 1 - n.$$

Then

$$F_0 \ge \tfrac{1}{10}n^3(1-n) + \tfrac{3}{5}t^4 \ge \mathcal{H}/10.$$

On $\mathcal{D}_+$, $G_0 \ge 2\mathcal{H}/5$. Consequently, under the source's potential collection identity and spectral-shape bound,

$$\mathcal{I}_0 \le -\Theta^2/c - \mathcal{H}/(10c). \tag{14}$$

The continuous nonnegative function $\mathcal{H}$ vanishes exactly at $(v, b) = (0, 0)$ and $(1, 1)$.

Proof. For $n < 1$ put $z = t^2/(1-n)^2 \in [0,1]$. The source expands the even polynomial $F_0(1-n+t, 1-n-t)$ in degree-four Bernstein form in z. Its five coefficients $h_i(n)$ have the replayed lower bounds

$$(h_0, h_1, h_2, h_3, h_4)/(1-n) \geq (3n^3, 31n, 1/5, 12/5, 9/5).$$

For precision, they are obtained from $F_0 = \sum_{j=0}^{4} f_j(n) t^{2j}$ by

$$h_i(n) = \sum_{j=0}^{i} \frac{\binom{i}{j}}{\binom{4}{j}} f_j(n)(1-n)^{2j}.$$

After dividing by $n^3(1-n)$, $n(1-n)$, or $1-n$ as appropriate, the resulting polynomials have the stated nonnegative lower bounds on each of four quarter intervals. These are the twenty potential-quotient groups in the exact replay.

The degree-four coefficients of $n^3/10 + (3/5)z^2$ are

$$n^3/10 + (3/5)i(i-1)/12 \quad (0 \leq i \leq 4).$$

For $i = 0$ the comparison is immediate; for $i = 1$ use $n^3 \leq n$; for $i = 2, 3, 4$ the candidate upper bounds are $1/5$, $2/5$, $7/10$, respectively, no larger than the three available constants. Thus

$$F_0 \geq (1-n)\{n^3/10 + (3/5)z^2\} \geq n^3(1-n)/10 + 3t^4/5.$$

The endpoint $n = 1$ contains only $t = 0$ and follows by continuity, without division there. Since $n^3 \geq n^4$, this dominates $\mathcal{H}/10$. On $\mathcal{D}_+$, $1-b = n+t$, $t \geq 0$, and (11) gives $G_0 \geq (2/5)(n+t)^4 \geq 2\mathcal{H}/5$.

The source identity is

$$\mathcal{I}_0 = -E_0 - 6v^2(1-\sigma)L_v - 6b^2(1-\bar{\sigma})L_b.$$

Here $L_v \geq 0$, and $0 \leq b^2(1-\bar{\sigma}) \leq 2\delta^2$. If $L_b \geq 0$, the upper bound is $-E_0 = -(\Theta^2 + F_0)/c$. If $L_b < 0$, it is $-E_0 - 12\delta^2 L_b = -(\Theta^2 + G_0)/c$. Both give (14); exchange labels on the other ordered half. Zero norms use the source's continuous potential products. Finally, $\mathcal{H} = 0$ forces $t = 0$ and $n \in \{0, 1\}$, exactly the two listed points. □

Corollary 5.6 (Certificate-based terminal reduction)**.** *Assume continuous norms on a connected manifold, the source's integral sign* $\mathbf{E}\mathcal{I} \geq 0$*, its quartic and quadratic comparisons, and* $r^2 \leq j$*. Assume its potential collection and shape bounds as in Proposition 5.5. Then the spectral map* (v, b) *is constant with value* $(0,0)$ *or* $(1,1)$*. If both blocks are assumed nonzero and the sectional curvature is strictly positive, these hypotheses are inconsistent.*

Proof. The completed square together with (14) gives

$$0 \leq \mathbf{E}\mathcal{I} \leq -\mathbf{E}\left[(\Theta - cr)^2/c + c(j - r^2) + \mathcal{H}/(10c)\right] \leq 0.$$

Thus $\mathcal{H} = 0$ almost everywhere. It is continuous and every nonempty open set has positive Riemannian volume, so it vanishes everywhere. The continuous image of a connected space is connected; a subset of the two distinct points $\{(0,0), (1,1)\}$ is connected only if it is a singleton. The first point makes both blocks identically zero, contrary to their assumed nonvanishing. The second has $v + b = 2$, contrary to $v + b \leq x + y < 2$. □

This replaces the source's final active-gradient/level-set strictness step by an explicit potential certificate. It retains the earlier weak-norm, Hessian, moment-balance, and finite-algebraic inputs. It neither avoids their verification nor proves the classification independently. In nonnegative geometry the point $(1,1)$ is not excluded by this last inequality; identifying its equality geometry belongs to a different problem.

6 Quantitative Residual Closure

This section proves a conditional extension of the final square argument, not an approximate-Einstein classification. All expectations use a probability volume measure. On a closed connected Riemannian manifold assume the scalar Poincare inequality

$$\mathrm{Var}(f) \le \tfrac{1}{4}\mathbf{E}|df|^2 \quad (f \in H^1). \tag{15}$$

It follows from (4) in the source setting and is also the normalized Lichnerowicz bound. Let nonnegative Lipschitz functions v, b satisfy $v + b \le 2$. Quotients with a zero function value use the zero-gradient convention, and assume $j \in L^1$; this holds for the source norms by (3) and bounded derivatives. Define r, j, c as in (3.4); Θ may be any bounded measurable real function.

Lemma 6.1 (Exact rational deficit)**.** *For $s > 0$, writing $r_s = 1 + 6s + s^2$, one has*

$$\frac{1}{s} - R(s)^2 = \frac{(s-1)^4}{s r_s^2}, \qquad R(s) \le s^{-1/2}. \tag{16}$$

For $A = |\nabla v|^2$, $B = |\nabla b|^2$, and $v, b > 0$,

$$\begin{aligned} j - r^2 = {} & A^2 \frac{(v-1)^4}{v r_v^2} + B^2 \frac{(b-1)^4}{b r_b^2} \\ & + AB\left[(v^{-1/2} - b^{-1/2})^2 + 2\{(vb)^{-1/2} - R(v)R(b)\}\right]. \end{aligned} \tag{17}$$

Every term on the right is nonnegative. For

$$F(s) = \tfrac{1}{2}(s-1)|s-1|,$$

the zero-set extension satisfies almost everywhere

$$j - r^2 \ge \frac{1}{578}\left(|\nabla F(v)|^4 + |\nabla F(b)|^4\right). \tag{18}$$

Proof. Expanding $r_s^2 - 16s(1+s)^2$ gives $(s-1)^4$. Since $R(s) > 0$, this proves the first line and its inequality. Expansion of the two quadratic expressions in A, B proves (17). Its last brace is nonnegative by the first line. On $0 < s \le 2$, $s r_s^2 \le 2 \cdot 17^2 = 578$. Also $F'(s) = |s-1|$, so the chain rule gives (18). If a norm is zero, its gradient vanishes almost everywhere there; delete its terms before taking the limit, leaving the same conclusion. F is continuously differentiable with bounded derivative on $[0, 2]$, so these compositions belong to H^1 without any singular chain rule. □

Theorem 6.2 (Residual-to-variance and two-block obstruction)**.** *Suppose $\mathcal{I}$ and a nonnegative a_+ are integrable and $a_- \ge 0$ is a number, with*

$$\mathbf{E}\mathcal{I} \ge -a_-, \qquad \mathcal{I} \le -\frac{(\Theta - cr)^2}{c} - c(j - r^2) + a_+ \quad a.e.$$

Set $\varepsilon = a_- + \mathbf{E}a_+$. Then

$$\mathbf{E}\left[\frac{(\Theta - cr)^2}{c} + c(j - r^2)\right] \le \varepsilon, \qquad \mathbf{E}\left(|\nabla F(v)|^4 + |\nabla F(b)|^4\right) \le 5780\varepsilon. \tag{19}$$

If (15) holds, then

$$\mathrm{Var}(v) + \mathrm{Var}(b) \le 2^{5/4}(5780\varepsilon)^{1/4}. \tag{20}$$

If additionally $\mathbf{E}v^2 \ge 1$, $\mathbf{E}b^2 \ge 1$ and $v + b \le 2 - \zeta$ for some $\zeta > 0$, necessarily

$$\varepsilon \ge \frac{\zeta^4}{184960}. \tag{21}$$

Proof. Integration and subtraction give the first inequality in (19); Lemma 6.1 and $c \geq 1/10$ give the second. For $u, w \in [0, 2]$,

$$|u - w|^2 \leq 4|F(u) - F(w)|.$$

On one side of 1 this follows from the difference of two squares, even with constant two. On opposite sides put $\alpha = u - 1 \geq 0$, $\beta = 1 - w \geq 0$ and use $(\alpha + \beta)^2 \leq 2(\alpha^2 + \beta^2)$. Integrating against two independent copies of $d\nu$, and applying Cauchy–Schwarz, yields

$$\operatorname{Var}(v) = \tfrac{1}{2}\mathbf{E}\mathbf{E}'|v - v'|^2 \leq 2\sqrt{2\operatorname{Var}(F(v))} \leq \sqrt{2}\big(\mathbf{E}|\nabla F(v)|^4\big)^{1/4}.$$

The last inequality uses (15) and $\mathbf{E}|df|^2 \leq (\mathbf{E}|df|^4)^{1/2}$. Apply the same reasoning to b, then $a^{1/4} + d^{1/4} \leq 2^{3/4}(a + d)^{1/4}$ to obtain (20).

Write $\mu = \mathbf{E}v \geq 0$, $V = \operatorname{Var}(v)$. The second-moment assumption gives $\mu^2 \geq 1 - V$. If $0 \leq V \leq 1$, then $\mu \geq \sqrt{1 - V} \geq 1 - V$; if $V \geq 1$, the last inequality is automatic. Consequently

$$2 - \zeta \geq \mathbf{E}(v + b) \geq 2 - \operatorname{Var}(v) - \operatorname{Var}(b).$$

Combine with (20) and raise to the fourth power: $\zeta^4 \leq 32 \cdot 5780\varepsilon$. This is (21). □

For a fixed compact strictly positive geometry, $\zeta = \min_M(2 - x - y) > 0$ is available and $v + b \leq 2 - \zeta$. In the exact balanced source argument $\varepsilon = 0$, which contradicts two nonzero blocks. The theorem quantifies the tolerance needed at that final step; it does not provide a uniform ζ over all strictly positive metrics, a distance to the Einstein equation, or a curvature-tensor closeness theorem. If either moment gap is absent, the last conclusion does not follow.

Corollary 6.3 (Stronger closure under a strict spectral margin)**.** *Retain the integral lower bound and other hypotheses of Theorem 6.2, and assume the strengthened pointwise bound supplied by Theorem 5.4. Put*

$$W = \frac{(1 - \min(v, b))^2}{2720}(|\nabla v|^2 + |\nabla b|^2).$$

Then

$$\mathbf{E}\left[\frac{(\Theta - cr)^2}{c} + c(j - r^2) + W\right] \leq \varepsilon. \tag{22}$$

If $v + b \leq 2 - \zeta$, $\zeta > 0$, then

$$\mathbf{E}(|\nabla v|^2 + |\nabla b|^2) \leq 10880\varepsilon/\zeta^2, \qquad \operatorname{Var}(v) + \operatorname{Var}(b) \leq 2720\varepsilon/\zeta^2.$$

Under both unit second-moment gaps, necessarily

$$\varepsilon \geq \zeta^3/2720. \tag{23}$$

Proof. Add the retained negative quadratic term before integrating. Since $\min(v, b) \leq (v + b)/2 \leq 1 - \zeta/2$, the weight in W is at least $\zeta^2/10880$. This gives the energy bound. Apply (15) to both norms, then the already proved $\zeta \leq \operatorname{Var}(v) + \operatorname{Var}(b)$. Rearrangement proves (23). □

This improvement is structural: the new term controls a quadratic norm-gradient energy, whereas (18) controls a degenerate quartic energy after transformation. It is not merely a larger coefficient in the same inequality. On $0 < \zeta \leq 2$, (23) is stronger than (21). Neither is asserted to have an optimal constant or exponent under stronger hypotheses.

Corollary 6.4 (Spectral concentration with a potential residual)**.** *If the same integral residual budget also includes* (14)*, then*

$$\mathbf{E}\left[(\Theta - cr)^2/c + c(j - r^2) + W + \mathcal{H}/(10c)\right] \le \varepsilon, \qquad \mathbf{E}\mathcal{H} \le 4\varepsilon.$$

For $0 < a \le 1/2$*,*

$$\nu\{a \le n \le 1 - a\} \le 4\varepsilon/a^5, \qquad \nu\{|t| \ge a\} \le 4\varepsilon/a^4.$$

Proof. All four summands are nonnegative and $c \le 2/5$, so $\mathcal{H} \le 4\mathcal{H}/(10c)$. On the first event $n^4(1-n) \ge a^5$; on the second $t^4 \ge a^4$. Integrate these lower bounds. □

These are measure bounds for Weyl-norm spectral variables. They are not Gromov–Hausdorff, diffeomorphism, or tensor-parallelness estimates.

6.1 Moment imbalance as an explicit cost

Proposition 6.5 (Unbalanced variance addition)**.** *Assume* (4)*,* $v, b \ge 0$*, and* $v + b \le 2$*. Put*

$$d = v - b, \quad e_2 = \mathbf{E}(v^2 - b^2), \quad x_0 = \mathbf{E}(\delta d), \qquad V_D = D\left[|\nabla d|^2/K - (1 - \delta^2/4)d^2\right].$$

Then

$$\begin{aligned} \mathbf{E}V_D &\ge \frac{D}{4}\{\mathrm{Var}(\delta d) - 2e_2x_0 - e_2^2\} \\ &\ge -\eta_{\mathrm{bal}}, \qquad \eta_{\mathrm{bal}} = \frac{D}{4}(2|e_2| + e_2^2). \end{aligned} \tag{24}$$

Thus, if all other source integral signs and pointwise bounds hold with integrated error ε_0*, Theorem 6.2 holds with total error* $\varepsilon_0 + \eta_{\mathrm{bal}}$*. A sharper permissible cost is*

$$\frac{D}{4}\max\{0,\, 2e_2x_0 + e_2^2 - \mathrm{Var}(\delta d)\}.$$

Proof. Since $(2 - \delta)d = v^2 - b^2$, one has $\mathbf{E}d = (x_0 + e_2)/2$. Apply the variance bound to d to find

$$\mathbf{E}V_D \ge D\{\tfrac{1}{4}\mathbf{E}[(\delta d)^2] - (\mathbf{E}d)^2\}.$$

Expansion proves the first line. If $s = v + b \in [0, 2]$, then $|d| \le s$ and $|\delta d| \le (2 - s)s \le 1$, so $|x_0| \le 1$. Discard the nonnegative variance and take absolute values to obtain the second line. The sharper expression simply retains the signed first line, and is nonnegative by construction. □

The source's weighted Hessian inequality does not itself require moment balance. The quartic, quadratic, and potential expressions use the same fixed weights independently of that balance. Only the variance addition changes its integral sign, which is why this conditional replacement is well-defined. It is not legitimate to use it on an arbitrary non-Einstein metric while importing the source's differential identities unchanged. At the equality boundary $\zeta = 0$, (21) supplies no exclusion, precisely as it should.

7 Primary Literature and Comparison

Public searches used mathematical topics and public titles only. The following primary literature was checked during the October 7/8, 2026 session; the status below concerns the statements retrieved, not a claim to exhaustive independent peer review.

Gursky–LeBrun [2] classifies compact oriented Einstein four-manifolds with positive intersection form and nonnegative sectional curvature as Fubini–Study $\mathbb{CP}^2$, up to rescaling and isometry. Positive intersection form is a hypothesis, not a consequence of nonnegative sectional curvature. Its classical Weyl-gap framework is relevant here. Cao–Tran [3] treats additional sectional-curvature pinching. These results do not automatically discharge an unrestricted two-block coupling argument.

Cheng's September 20, 2026 preprint [4] proves the positively curved oriented conclusion with $\chi \leq 3$, including a consequence for a nontrivial Killing field. Gursky–Malchiodi's September 25, 2026 preprint [5], Theorem 1.1, assumes $2\chi - 3|\tau| \leq 4$ and uses conformal Green functions and mass bounds. The latter manuscript explicitly retains the topological restriction and identifies its removal as conjectural. Their public statements therefore cannot be cited as proofs of the unrestricted M671 claim.

Catino–Dameno [6] assumes simplicity or a uniform separation hypothesis on the largest self-dual Weyl eigenvalue, deriving conformal Kahler geometry after at most a double cover. Di Cerbo's corrected July 22, 2026 version [7] classifies saturation of the Weyl-norm bound under a fixed sign of sectional curvature. These distinguish spectral-shape and equality phenomena from M671's strictly positive interior. In particular, a nonnegative boundary model is not a counterexample to a theorem with strict sectional positivity, and a proof about that boundary does not prove the interior classification.

Ozuch's October 6, 2026 preprint [8] announces positive *scalar*-curvature Einstein metrics on $\#_k(S^2 \times S^2)$ and $k\mathbb{CP}^2\#(2k+1)\overline{\mathbb{CP}}^2$ for $k \geq 2$. Its abstract was inspected for this statement, not its construction proof, and this very recent preprint is not treated as externally peer-reviewed here. The statement emphasizes why positive Einstein constant, equivalently positive scalar curvature for Einstein metrics, cannot substitute for positive sectional curvature in M671. It does not contradict the strict-sectional claim.

Roy's primary positivity report [9] and the classical Bernstein literature explain the coefficient method; that basis is not a new Ram invention. The local improvements in Sections 5 and 6 use standard convexity and energy tools but derive specialized consequences for the exact M671 data. Historical novelty beyond this local comparison has not been established.

8 Verification, Advantages, and Limits

The finite local checker verifies the rational deficit identity, the source's physical-metric multiplication identity, the imbalance expansion, the constant $184960 = 32 \cdot 5780$, finite rational examples of the deficit inequality, and the combinatorial support bound in (10). These are bounded tests of the displayed local mathematics, not random numerical evidence for global geometry. The proof of each theorem above is independent of those sample tests.

The executed local check passed nine groups, including 2,049 norm-deficit witnesses, 231 barycentric witnesses, twenty coefficients for the common potential minorant, 349 shifted nonzero matrix coefficients, and 110 scalar coefficients. It also checks all corresponding exceptional zero weights. Its matrix checks use rational principal minors, not floating-point eigensolvers. The symbolic checks used installed SymPy 1.11.1; no software was installed.

The exact scalar and matrix coefficient replay is a distinct verification layer. Forty groups

passed: eight interval groups for the quartic scalar signs, nine triangular matrix groups, three triangular scalar groups, and twenty interval groups for the five potential quotients. The even potential identities, all recorded zero exceptions, and the additional vanishing-factor margins were also checked over $\mathbb{Q}$. This count concerns Section 7; it does not include the moment-obstruction and rational-integral arithmetic addressed by the separate replay below. No claim to execute the source's missing packaged checker is made.

A separate bounded exact-arithmetic replay, recorded in `_audit/M669-M671-extra-finite-20261008/results.json`, reports 120 checks passed with no failures or exceptions. Its script and read scope were inspected. It reconstructs all 127 coefficients of seven targeted moment Bernstein tables: the degree-eight L table, four degree-seventeen tables for the source's univariate derivative certificate, and both degree-twenty-two endpoint polynomials. It checks their printed minima, affine reconstructions, third-derivative identities, and endpoint-weight identities. It also evaluates all three upper rational-integral rows and both M671 lower rows, including the base term and totals. The printed upper row sum is $40781/500000 > 163/2000 > 2/25$; the lower base-plus-row bound is $1007539/750000 > 1343/1000 > 4/3$. Four source-file hashes were unchanged within that replay. This is attributed computational evidence, not a claim of a second execution here.

These finite checks establish only the recorded algebraic identities, coefficient signs, and rational inequalities. They do not verify the analytic sine/Taylor comparisons or brackets for π, the probability-measure separation and support argument, the polar comparison theorems, the bundle calculations, or the global isometry assembly. The bounded independent source review located no substantive defect in its scope, but neither that conclusion nor the arithmetic passes constitutes unconditional theorem verification.

Advantages. Exact rational representations prevent tolerance ambiguities and retain source-specific boundary zeros. The active-coefficient profile gives a usable perturbation budget where a uniform eigenvalue margin is impossible. The residual theorem converts a qualitative final sign into a quantitative necessary condition and accommodates measured moment imbalance without silently changing the equations.

Costs and limitations. Bernstein positivity is sufficient, not necessary at a fixed degree; subdivision may increase data size. The coefficient discovery procedure has no asserted global optimality or runtime bound. Preserving a face requires exact identities rather than numerical rank guesses. The residual constant is conservative and is not described as sharp. Fourth-order degeneracy at $v = 1$ limits direct norm-gradient coercivity, and the transformed-variable proof does not remove that degeneracy. All geometric errors must actually be bounded before insertion into ε_0.

Research directions. A useful next step is to certify the complete map from tensor identities to finite arrays, with independent conventions and zero-set treatment. Another is to derive natural error norms for approximate differential identities and determine whether they yield the integrable a_+ and a_- assumed here. Exact boundary-constrained coefficient optimization may improve the final tolerance, but a better number alone is not a new geometric theorem. Quantitative curvature pinching or metric stability would require additional compactness and regularity estimates beyond this report.

A Exact Local Checks and Reproduction Scope

The companion local script is `_audit/M671-author-checks.py`; its machine-readable output is `_audit/M671-author-check-results.json`. The read-scope and input hashes are in `_audit/M671-author-inputs.json`. They are local reproducibility records.

For the physical-metric check, the normalized form matrices are

$$\frac{\mathsf{B}}{v} = \begin{pmatrix} 4\sigma/3 & -4\rho/\sqrt{3} & 0 \\ -4\rho/\sqrt{3} & -4\sigma & -1 \\ 0 & -1 & 0 \end{pmatrix}, \quad H = \operatorname{diag}(5/3, 5, 1).$$

Direct multiplication using $\sigma^2 + \rho^2 = 1$ gives

$$\frac{\mathsf{B}H^{-1}\mathsf{B}}{v^2} = \begin{pmatrix} 16/15 & 0 & 4\rho/(5\sqrt{3}) \\ 0 & 21/5 & 4\sigma/5 \\ 4\rho/(5\sqrt{3}) & 4\sigma/5 & 1/5 \end{pmatrix}.$$

The identity checks a finite local interface used by the source, not its covariant derivative equation. The bounded samples include zero gradients at zero norms and exact values on the axial line; they do not sample away a singular coefficient.

References

[1] OpenAI, *Positively curved Einstein four-manifolds*, September 23, 2026, pinned local source at commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`. Public locator: https://github.com/openai/math/tree/adc7f1241b42e322a6451854ab7e4b4c146bf78a/preprints/Positively-curved-Einstein-four-manifolds-September-23-2026.

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